Aniket Rege

Department of Computer Sciences
University of Wisconsin-Madison
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Google Scholar
Personal Website

Education

University of Wisconsin-Madison *2023 - present*
PhD in Computer Sciences
Advisor: Prof. Ramya Korlakai Vinayak & Prof. Yong Jae Lee

University of Washington *2021 - 2023*
MS in Electrical and Computer Engineering
Advisor: Prof. Ali Farhadi

National Institute of Technology Karnataka *2014 - 2018*
B.Tech in Electronics and Communication Engineering
Advisor: Prof. Muralidhar Kulkarni

Work Experience

Meta Reality Labs, Redmond *May - Dec, 2025*
Research Scientist Intern, Reality Labs
Advisors: Dr. Hyo Jin Kim & Dr. Yuning Chai

NVIDIA, Redmond *June - Sept, 2022*
Applied Research Intern, Deep Learning for Autonomous Vehicles
Advisors: Dr. Nikolai Smolyanskiy & Dr. Stan Birchfield

Samsung Research India, Bangalore *2018 - 2021*
AI Camera Research Engineer, Visual Intelligence Group
Advisors: Sudha Velusamy & Dr. Shankar Venkatesan

Publications

Preprints

* equal contribution, † equal advising

1. **Sparks of Cooperative Reasoning: LLMs as Strategic Hanabi Agents.**
   Mahesh Ramesh, Aswinkumar Ramkumar, Pavan Thodima, Kaousheik Jayakumar,
   Aniket Rege†, and Manolis Vlatakis†.
   Under review @ ***ICML****, 2026.*
   - Knowledge-Intensive Multimodal Reasoning Workshop @ ICCV, 2025.
   - Bridging Language, Agent, and World Models Workshop @ NeurIPS, 2025
   - Scaling Environments for Agents Workshop @ NeurIPS, 2025
   - Multi-Turn Interactions in Large Language Models Workshop @ NeurIPS, 2025

Conference Publications

8. **Agentic Very Long Video Understanding.**
   Aniket Rege, Arka Sadhu, Yuliang Li, Kejie Li, Ramya Korlakai Vinayak, Yuning Chai,
   Yong Jae Lee, and Hyo Jin Kim
   *The 64th Annual Meeting of the Association for Computational Linguistics (****ACL****), 2026.*

7. **CuRe: Cultural Gaps in the Long Tail of Text-to-Image Models**.
   Aniket Rege, Zinnia Nie, Mahesh Ramesh, Unmesh Raskar, Zhuoran Yu, Aditya Kusupati†,
   Yong Jae Lee†, and Ramya Korlakai Vinayak†.
   *IEEE/CVF International Conference on Computer Vision (****ICCV****), 2025.*
   - Demographic Diversity in Computer Vision workshop @ CVPR, 2025 (**Oral**).
   - Critical Evaluation of Generative Models and Their Impact on Society Workshop @ ICCV, 2025 (**Oral**).
   - What is Next in Multimodal Foundation Models? Workshop @ ICCV, 2025.
   - Safe and Trustworthy Multimodal AI Systems Workshop @ ICCV, 2025.

6. **PAL: Sample-Efficient Personalized Reward Modeling for Pluralistic Alignment**.
Daiwei Chen, Yi Chen, **Aniket Rege**, Zhi Wang, and Ramya Korlakai Vinayak.
*International Conference on Learning Representations (**ICLR**), 2025.*
- Models of Human Feedback for AI Alignment workshop @ ICML, 2024 (**Oral**).
- Workshop on Behavioral Machine Learning @ NeurIPS, 2024 (**Spotlight**).
- Pluralistic Alignment workshop @ NeurIPS, 2024.

5. **AdANNS: A Framework for Adaptive Semantic Search**.
Aniket Rege*, Aditya Kusupati*, Sharan Ranjit S, Alan Fan, Qingqing Cao, Sham Kakade, Prateek Jain, and Ali Farhadi.
*Neural Information Processing Systems (**NeurIPS**), 2023.*
- Practical ML for Developing Countries Workshop @ ICLR, 2023 (**Oral**).

4. **Matryoshka Representation Learning**.
Aniket Rege*, Aditya Kusupati*, Gantavya Bhatt*, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi.
*Neural Information Processing Systems (**NeurIPS**), 2022.*
- Vision Transformers: Theory and Applications workshop @ NeurIPS, 2022 (**Oral**).
- Self-Supervised Learning - Theory and Practice workshop @ NeurIPS, 2022.
- Computer Vision in the Wild workshop @ ECCV, 2022.

3. **Spatio-Temporal Video Representation Learning for AI Based Video Playback Style Prediction**.
Rishubh Parihar*, Gaurav Ramola*, Ranajit Saha, Raviprasad Kini, **Aniket Rege**, and Sudha Velusamy.
*Structured Representations for Video Understanding workshop @ **ICCV**, 2021.*

2. **FabSoften: Face beautification via dynamic skin smoothing, guided feathering, and texture restoration**.
Sudha Velusamy, Rishubh Parihar, Raviprasad Kini, **Aniket Rege**.
*IEEE/CVF Conference on Computer Vision and Pattern Recognition workshops (**CVPR W**), 2020.*
- New Trends in Image Restoration workshop @ CVPR, 2020.

1. **QUIC Protocol Performance in Wireless Networks**.
Prashant Kharat*, **Aniket Rege***, Aneesh Goel*, and Muralidhar Kulkarni.
*IEEE International Conference on Communication and Signal Processing (**ICCSP**), 2018.*

Patents

3. **Method and electronic device for detecting candid moment in image frame**.
Aniket Rege*, Gaurav Ramola*, Nikhar Maheshwari*, Sudha Velusamy, Girish Kulkarni, Sai Pranav Mathivanan, Swadha Jaiswal, and Pradeep Kumar Vagireddi.
U.S. Patent 12,299,959, issued May 13, 2025.

2. **Method and electronic device for processing facial images**.
Sudha Velusamy, Rahul Varna, Raviprasad Kini, **Aniket Rege**, Rishubh Parihar, Daeyoung Hyun.
U.S. Patent 11,335,122, issued May 17, 2022.

1. **Mobile Accessory For 3D Selfie Creation**.
Aniket Rege*, Shankar Venkatesan*, Ankit Jalan*, Mynepalli Siva Chaitanya, Arko Sabui, Rishi Franklin, Viswanath Veera, Abhijeet Singh.
IN 411,097, issued March 23, 2018.

Theses

2. **Congestion Control in Wireless Networks**.
Aniket Rege*, Aneesh Goel*, and Muralidhar Kulkarni.
Undergraduate Thesis, Electronics and Communication Engineering, NITK Surathkal, 2018.

1. **MRL-AdANNS: Matryoshka Representation Learning for Web-Scale Adaptive Semantic Search**
Aniket Rege
Masters Thesis, Electrical and Computer Engineering, University of Washington, 2023.

Selected Awards and Honors

- **CS Departmental Scholarship**, UW-Madison *2023 - present*
- **NeurIPS Scholar Award** *2022*

- **University of Washington Graduate School Award** for NeurIPS '22 *2022*
- **Samsung HQ Best Project Award** as founding member of **Single Take Camera**, the USP of the Galaxy S20 smartphone *2021*
- **Samsung Best Paper Award '19 Finalist** for FabSoften, later published at CVPR '20, selected from all worldwide offices *2019*
- **Samsung India Outstanding Contribution Award** for shipping **AI Beauty Solution** to 50M+ smartphones worldwide *2019*
- **Road to Global Entrepreneurship Summit (GES) '17 national finalist** for *An Analysis of Sleep Hygiene*. Conducted by NITI Aayog under the Government of India. *2017*

Professional Responsibilities

- *Reviewing*: ICCV/ECCV (2023 - present), NeurIPS (2024 - present), ICLR (2024 - present), ICML (2025 - present), CVPR (2025 - present)
- *Workshop Organization*
  - Multimodal Alignment for a Pluralistic Society (MAPS) *CVPR 2026*
- *Graduate Teaching Assistant*, University of Wisconsin-Madison
  - CS 540: Introduction to Artificial Intelligence - Prof. Yudong Chen, Prof. Blerina Gkotse *Fall 2023-25*
- *Graduate Teaching Assistant*, University of Washington
  - CSE 493: Deep Learning - Prof. Ranjay Krishna *Spring 2023*
  - CSE 547: Machine Learning for Big Data - Prof. Tim Althoff *Winter 2023*
  - EE 596: Computer Vision: Classical and Deep Methods - Dr. Stan Birchfield *Fall 2021*
- *Mentorship*
  - Mahesh Ramesh
    MS CS, UW Madison *Jan '25 - May '26*
  - Unmesh Raskar
    MS ECE, UW Madison → ML Engineer @ Cadence *Jan '25 - May '25*
  - Zinnia Nie
    BS Student, UW Madison CS → MS Student @ U Michigan CS *Jun '24 - May '25*
  - Sharan Ranjit
    MS Student, UW ECE → Machine Learning Engineer @ Autodesk *Jun '22 - May '23*
  - Ankit Pandey
    Intern, Samsung R&D *Apr - Jul 2021*
  - Nikhar Maheshwari
    Intern, Samsung R&D → MS Student @ Stony Brook CS *May - Jul 2020*
- *Samsung University Representative* - Technical liason for hiring of students from NITK *2018 - 2021*
- *Undergraduate Teaching Assistantship, Linear Signals and Systems* - Department of ECE, NITK *2015 - 2016*

Extracurriculars

Convener and Head, Indian Society for Technical Education, NITK *Apr 2017 - May 2018*
Advisor: Dr. Basavaraju Manu, *NITK*

- Led a team of 100+ students to conduct technical talk and cross-domain technical projects across STEM fields, business, literature, and graphic design.
- Head of the Electrical Engineering special interest group.
- Initiated and organized the university's first cross-campus hackathon during the annual tech fest with SRIB.
- Pioneered NITK's first annual technical club project expo in association with IEEE-NITK.
- Initiated the formation of the club's alumni network on LinkedIn, as well as the club's first summer mentorship program.

Joint Convener and Head, Engineer (Technical Fest) *Aug - Oct 2017*

- Joint head of NITK's annual technical fest, Engineer '17.
- Co-led the Technites committee, a three day expo of EECS projects primarily focused on embedded electronics.

Performing Artist (Vocals), NITK Music Club *Aug 2016 - May 2018*